

ABHINANDAN KAUSTHUB

Aerospace Structures Engineer | Stress · Fatigue & Damage Tolerance · Composites

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PROFILE

Aerospace engineer specialising in stress and structural integrity, with a focus on fatigue, fracture and damage tolerance. What pulls me in is the whole chain: how a material behaves, how load flows through a structure, where a fatigue crack starts and when fracture takes over, because that is where structural integrity is really won or lost. I build and validate my own analysis tools alongside FEA, CFD and hands-on lab work. MS from Auburn University; raised in Singapore and educated across Singapore, Indonesia, India and the US.

EDUCATION

Master of Science, Aerospace Engineering | Auburn University (R1), Auburn, USA

Graduated May 2026

Coursework: Fatigue & Damage Tolerance · Composite Materials · Experimental Mechanics · Viscoelasticity · Deformation and Failure of Engineering Materials · Lightweight Space Structures · Phase Transformations in Materials · Biomaterials · Quantum Physics · Fluid Dynamics

Bachelor of Technology, Aeronautical Engineering | Manipal Institute of Technology, India

Graduated 2023

Coursework: Aircraft Design · Structural Mechanics · Finite Element Methods · Propulsion · Aerodynamics · Aerospace Materials

International schooling — Singapore (where I grew up) and Indonesia, before university in India and graduate school in the US.

TECHNICAL PROJECTS

Structural Integrity & Fatigue/Fracture Toolkit | Python, Streamlit

GitHub repo ↗

- Built a damage-tolerance app that grows a crack from initial to critical length with Paris, Walker and Forman laws and stops at $K_{\max} = K_{Ic}$. Predicted life for an edge-cracked SAE 1020 plate: **128,697 cycles vs. 128,792 published (-0.07%)**.
- Rainflow counter reproduces the **ASTM E1049** worked example exactly; feeds Miner's-rule damage with Morrow/SWT mean-stress correction, Goodman/Gerber/Soderberg diagrams and Monte Carlo life scatter across an aerospace alloy database (2024-T3, 7075-T6, Ti-6Al-4V, Inconel 718, 4340).
- Audited my own first release against the theory, found and fixed real modelling errors (fixed-step crack integration, Basquin range vs. amplitude, Goodman using S_u instead of S_e), then locked the physics in with **23 automated tests**.

Composite Laminate Analysis Toolkit | Python, Streamlit

GitHub repo ↗

- CLT solver for any stacking sequence (e.g. [0/45/-45/90]s) returning the ABD matrix and ply stresses in laminate and fibre axes. Checked against Kaw's textbook: **A_{11} +0.4%, E_x +0.5%, Tsai-Wu first-ply failure -0.05%**.
- Progressive ply-discount failure (separate matrix and fibre degradation) reporting first-ply and ultimate load, an 8-ply symmetric layup optimiser and a one-click PDF stress report; 8 materials incl. IM7/8552, T300/5208, AS4/3501-6 and Kevlar 49; **15 automated tests**.

Humidity Aging of Magnetorheological Elastomers | BTech capstone, team of 3 | LabVIEW, CATIA, AutoCAD

Manipal, 2023

- Cast silicone and 27 vol.% carbonyl-iron specimens, and designed an aluminium clamping fixture that we cut on an abrasive-waterjet CNC, then drilled and tapped by hand to mount strips for vibration testing.
- Ran free-vibration tests (accelerometer + LabVIEW) and log-decrement analysis before and after two weeks in an environmental chamber: **damping ratio rose 15-21%** (ζ 0.086→0.104 silicone; 0.102→0.117 MRE) and **MRE natural frequency fell ~20%** (140→112 rad/s), traced to moisture uptake softening the matrix.

Hypersonic Missile CFD | ANSYS Fluent, CATIA V5

Brahmastra, 2023

- Simulated a missile at **Mach 10, 90 km altitude and 5° angle of attack** (k- ω , pressure far-field) to map pressure loading, Mach contours and shock cones.
- Built the geometry in CATIA V5, carved the body out of a fluid enclosure by Boolean subtraction and meshed it in ANSYS Workbench with refinement around the nose and fins; solved as an ideal gas with the energy equation on, in parallel.
- Tracked lift and drag coefficients, showed the angle of attack concentrating pressure on the lower nose cone and lower fins, and used section planes from nose to tail to confirm the body and fins stay inside the shock cone.

EXPERIENCE

- Graduate Engineering Trainee (Internship) | Brahmastra Aerospace & Defence

Aug 2023 – Sep 2023
- Modelled a missile in CATIA V5, meshed the fluid domain in ANSYS Workbench and ran a **Mach 10, 90 km, 5° angle-of-attack** simulation in Fluent (pressure far-field, k- ω , energy equation) to map pressure loading, Mach contours and shock-cone clearance over the body and fins.
 - Set up a titanium HPT blade model (NACA 2412 section, 250 mm span) in SpaceClaim/Mechanical under 700 N tangential and 400 N axial load at 4,252 rpm, and reviewed the first five mode shapes for resonance risk.
 - Produced CATIA and SolidWorks part models and GD&T-compliant drawings for design review.

- Product Engineer – Associate (Internship) | Gowra Aerospace Technologies

May 2022 – Jul 2022
- Followed aerospace components from raw material through machining to delivery, inspecting parts against GD&T drawings and tolerance requirements before sign-off.
 - Used FMEA and SPC control charts with the production and quality teams to flag process risks and keep critical dimensions in control.

TECHNICAL SKILLS

Stress & FEA	Finite Element Analysis (ANSYS Mechanical / Workbench), Stress/Strain Analysis, Linear Static & Modal Analysis, Thermal-Structural Analysis
Fatigue, Fracture & DT	Damage Tolerance Analysis, Fatigue Crack Growth (Paris, Walker, Forman), Stress Intensity Factors & K _{IC} Fracture Checks, S-N (Basquin) & Strain-Life (Coffin-Manson), Mean-Stress Correction (Goodman, Gerber, Morrow, SWT), Rainflow Counting, Miner's Rule, Monte Carlo Life Scatter
Composites	Classical Lamination Theory, ABD Stiffness Matrix, Ply-by-Ply Stress/Strain, Max Stress / Tsai-Hill / Tsai-Wu, Progressive Ply Failure, Stacking-Sequence Optimisation
Experimental	Free-Vibration Testing, Accelerometer DAQ (LabVIEW), Logarithmic-Decrement Damping, Environmental-Chamber Aging, Specimen & Fixture Fabrication
CFD	ANSYS Fluent, Compressible & Hypersonic Flow, k- ω Turbulence Modelling, Navier-Stokes
Programming	Python (NumPy, SciPy, Pandas, Matplotlib, Streamlit, ReportLab, pytest), MATLAB, Scientific Computing
CAD & Modelling	CATIA V5, SolidWorks, AutoCAD, ANSYS SpaceClaim, Parametric & Surface Modelling, GD&T, Engineering Drawings
Quality	FMEA, Statistical Process Control (SPC), Dimensional Inspection

CERTIFICATIONS

- Google Project Management Professional Certificate — Google
- Infor LN – Engineering Module — in progress